

Power BI & Dashboards

Turn Data Into Stunning Visual Stories

■ DATA VISUALIZATION

■ LEARNING OUTCOME

By the end of this PDF, you will build interactive Power BI dashboards that communicate insights clearly, using the right charts, slicers, and DAX formulas.

■ Skills You Will Gain

- Connect Power BI to data sources
- Build interactive dashboards with slicers
- Choose the right visualization for each insight
- Write basic DAX formulas for KPIs
- Design clean, professional dashboard layouts
- Publish and share reports

■ Power BI skills add ■1–5 LPA to your salary. Companies pay more for analysts who can visualize, not just calculate.

SECTION 1

Power BI Overview

Component	What It Does
Power Query (M Language)	Import, clean, and shape data before analysis
Data Model	Define relationships between tables (like SQL joins)
DAX (Data Analysis Expressions)	Write formulas for custom KPIs and calculations
Report Canvas	Build visual charts, tables, and slicers
Power BI Service	Publish and share dashboards online

SECTION 2

Connecting to Data

1	Open Power BI Desktop → Get Data → Choose source (Excel, SQL, Web, CSV)
2	Transform Power Query Editor opens → clean data: rename, filter, change types
3	Load Click 'Close & Apply' to load clean data into the model
4	Relate Model View → drag to connect tables (e.g., OrderID in Orders ↔ Products)
5	Build Go to Report View → drag fields to create visuals

SECTION 3

Choosing the Right Chart

Insight Goal	Best Chart	Power BI Visual
Compare categories	Bar / Column Chart	Clustered Bar Chart
Show trend over time	Line Chart	Line / Area Chart
Show proportions	Pie / Donut	Pie Chart (max 5 slices)
Show relationships	Scatter Plot	Scatter Chart

Geographic data	Map	Filled Map / Map visual
Show KPI	Card / Gauge	Card Visual or KPI Visual
Detailed data	Table / Matrix	Table or Matrix Visual

SECTION 4

Essential DAX Formulas

```
-- Total Sales (Measure) Total Sales = SUM(Sales[Amount]) -- Average Order Value Avg Order Value = AVERAGE(Orders[Amount]) -- Count distinct customers Unique Customers = DISTINCTCOUNT(Orders[CustomerID]) -- Year-over-Year Growth % YoY Growth = DIVIDE([This Year Sales] - [Last Year Sales], [Last Year Sales]) -- Running Total Running Total = CALCULATE([Total Sales], DATESYTD(Calendar[Date]))
```

SECTION 5

Dashboard Design Principles

Principle	Application
One Glance Insight	Put top KPI cards (Revenue, Orders, Profit) at top-left
5-Second Rule	Viewer should understand the main message in 5 seconds
Consistent Colors	Use max 3 colors: primary, secondary, alert (red for bad)
Hierarchy	Big numbers at top, details at bottom
White Space	Don't crowd — empty space improves readability
Mobile Layout	Use Power BI's Phone Layout for mobile viewers

SECTION 6

Slicers & Interactivity

- Slicer Visual: drag any dimension field (Date, Region, Product) → creates a filter
- Date Slicer: use 'Between' style for date range picker
- Drill-Through: right-click a data point → 'Drill through' to detailed page
- Cross-Filtering: clicking one visual automatically filters all others
- Bookmarks: save a specific filter state and link to buttons
- Tooltip Page: create a custom hover tooltip with more detail

■ PRACTICE QUESTIONS

Q1. What does DAX stand for and what is it used for?

Answer: Data Analysis Expressions — used to write custom calculations and KPIs in Power BI.

Q2. Which chart type is best for showing sales trends over 12 months?

Answer: Line Chart — it clearly shows trends and patterns over time.

Q3. What is the difference between a Measure and a Calculated Column in DAX?

Answer: Measure: calculated dynamically based on filter context. Calculated Column: computed row by row and stored in the table.

Q4. How do you connect two tables in the Power BI data model?

Answer: In Model View, drag the common key field from one table to the matching field in another.

Q5. Name 3 best practices for dashboard design.

Answer: 1) Put KPIs at top-left, 2) Use max 3 colors, 3) Keep each visual focused on one insight.

■ MINI PROJECT

Apply what you've learned with a real dataset!

Dataset: sales.xlsx (Date, Region, Product, Category, Units, Revenue, Cost)

Steps:

1	Import Connect Excel to Power BI, clean in Power Query
2	Model Create Date table, relate to Sales
3	Measures Total Revenue, Total Cost, Profit, Profit Margin %
4	Page 1 KPI Cards: Revenue, Orders, Profit + Line chart: Monthly trend
5	Page 2 Map: Revenue by Region + Bar: Top 10 Products
6	Slicers Add Year, Region, Category slicers for interactivity

Expected Output: A 2-page interactive Power BI dashboard (publish to Power BI Service)

Insights to Find: Best region, top product, profit trend, YoY growth comparison

■ Power BI / DAX Cheat Sheet	
<code>SUM(col)</code>	Sum all values
<code>AVERAGE(col)</code>	Mean value
<code>COUNT(col)</code>	Count non-blank rows
<code>DISTINCTCOUNT(col)</code>	Count unique values
<code>CALCULATE(expr, filter)</code>	Change filter context
<code>FILTER(tbl, condition)</code>	Return filtered table
<code>RELATED(col)</code>	Get value from related table
<code>DIVIDE(num, den, 0)</code>	Safe division (avoid errors)
<code>SUMX(tbl, expression)</code>	Iterate and sum per row
<code>DATESYTD(date_col)</code>	Year-to-date filter
<code>FORMAT(value, 'format')</code>	Number/date formatting

<code>IF(cond, true, false)</code>	Conditional measure
------------------------------------	---------------------